

15-06-2016 Esame Finale **Basi di Dati**

Surname/Cognome: _____.

Name/Nome: _____.

Matricola:

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e-mail: _____.

Instructions:

1. Il testo d'esame deve essere riconsegnato anche in caso di ritiro dall'esame.
2. Non è consentito rimuovere nessuna pagina da questo esame.
3. Solo le risposte scritte a penna verranno considerate.
4. Per la brutta copia può essere usato il retro delle pagine del testo e le pagine bianche alla fine del testo. Tutto ciò che viene scritto fuori dagli spazi per le risposte non viene valutato.

Marks

Q. 1

1.1 _____/6 pts

1.2 _____/2 pts

Q. 2

2.1 _____/2 pts

2.2 _____/2 pts

2.3 _____/2 pts

2.4 _____/2 pts

Q. 3 _____/3 pts

Q. 4

4.1 _____/5 pts

4.2 _____/5 pts

4.3 _____/6 pts

Q. 5 _____/3 pts

Q. 6 _____/3 pts

Q. 7 _____/3 pts

Q. 8 _____/4 pts

Q. 9 _____/2 pts

Q. 10 _____/2 pts

Q. 11 _____/4 pts

Q. 12 _____/4 pts

Total: _____ / 60 pts

Q. 1 (8 pts) Given the following CREATE TABLE statements :

```
CREATE TABLE Team (
  name CHAR(8) PRIMARY KEY,
  points INTEGER NOT NULL,
  home VARCHAR(250) NOT NULL UNIQUE,
  FOREIGN KEY (home) REFERENCES Stadium)
```

```
CREATE TABLE Stadium (
  name VARCHAR(250) PRIMARY KEY,
  city VARCHAR(10),
  capacity INTEGER )
```

```
CREATE TABLE Derby (
  team1 CHAR(8),
  team2 CHAR(8),
  date TIMESTAMP,
  onTV BOOLEAN,
  earnings DECIMAL(12,3),
  PRIMARY KEY (team1, team2, date),
  FOREIGN KEY (team1, team2, date) REFERENCES Match)
```

```
CREATE TABLE Match (
  team1 CHAR(8),
  team2 CHAR(8),
  date TIMESTAMP,
  stadium VARCHAR(250),
  tickets INTEGER,
  FOREIGN KEY (stadium) REFERENCES Stadium
  FOREIGN KEY (team1) REFERENCES Team
  FOREIGN KEY (team2) REFERENCES Team
  PRIMARY KEY (team1, team2, date))
```

1.1 (6 pts) Design the ER schema that produces the above definitions Non-standard notation (i.e, different from what present in the book and/or in the slides) will be considered a mistake.

1.2 (2 pts) Is it possible that there are more tuples in Team than in Stadium ?

A. ☐ YES B. ☐ NO

Q. 2 (8 pts) Given the following tables, and with only the tuples present below (consider Z empty!)

S(A, B, C)			T(D, E, F)			Z(G, H, F)			R(A, B, C)		
A	B	C	D	E	F	G	H	F	A	B	C
1	a	e	a	e	1				a	e	1
1	a	e	a	e	2				a	f	2
1	a	e	a	e	3				a	g	3
1	a	e	a	e	4				a	null	4

Taking into consideration only the tuples showed above, without any further assumption, answer the following:

2.1 (2 pts) Can (G, H, F) be a primary key for Z ?

A. ☐ YES B. ☐ NO

2.2 (2 pts) Is (D, F) a primary key for T ?

A. ☐ YES B. ☐ NO C. ☐ UNDECIDIBLE

2.3 (2 pts) Can (B, C, A) be a foreign key for S referencing T (assuming that (D, E, F) is a key for T)?

A. ☐ YES B. ☐ NO

2.4 (2 pts) Is (A, B) a key for R ?

A. ☐ YES B. ☐ NO C. ☐ UNDECIDIBLE

Q. 3 (3 pts) Consider the following schema instances (structure and tuples) :

S(A, B)	
A	B
1	10
2	20
3	30

T(B, C, D)		
B	C	D
1	10	NULL
3	NULL	w

U(D, E)	
D	E
NULL	f
f	w
m	NULL

Show the results of the following queries and write the results into a table format:

(a) `SELECT S.B+10, T.C+10, T.D FROM S JOIN T ON S.B*10 >= T.C*10`

Q. 4 (16 pts) A database stores information about user submitted recipes and ingredients:

Recipe (userID, code, title, description, username, email, **vegan**, **vegetarian**, difficulty)

Ingredient (code, name, description, **vegan**, **vegetarian**)

RecipeIngredient (userID, recipeCode, ingredientCode, recipeTitle, ingredientName, quantity)

Recipes are identified by userID and a recipe code. Then it stores title and description, the username and email of who wrote it, a boolean flag to mark it vegan and one for vegetarian, finally a difficulty value from 1 to 5. Each ingredient has a code, a name and a description, two boolean flags to mark it vegan and vegetarian. A recipe is made by one or more ingredient, and an ingredient can appear in many recipes. For each ingredient in each recipe the database stores the required quantity, the title of the recipe that uses it and the name of the ingredient.

userID, recipeCode in **RecipeIngredient** are foreign keys referencing **Recipe**; ingredientCode references **Ingredient**. Each user has a single email and username. If a recipe or an ingredient are vegan, then they are also vegetarian.

4.1 (5 pts) Provide in **RELATIONAL ALGEBRA** the query to return users that submitted only vegan recipes.

4.2 (5 pts) Provide in **RELATIONAL ALGEBRA** the hardest non-vegan recipe.

4.3 (6 pts) Provide in relational algebra the query that finds the codes of the ingredients that are used in all the recipes using only the operations $\pi, \sigma, \times, \bowtie, -, \cup, \cap$.

Q. 5 (3 pts) Consider the following relational schema:

Driver(did, name)

Car(cid, color)

Rent(did, cid, date)

The relationship Rent indicates that a driver has rented a car on a specific date. The underlined attributes indicate keys. The attribute Rent.cid is a Foreign Key to Driver and the Rent.cid is a foreign key to Car. Give the ER diagram of the specific relational schema

Q. 6 (3 pts) Consider a database that stores information about user submitted recipes and ingredients:

Recipe (userID, code, title, description, username, email, vegan, vegetarian, difficulty)

Ingredient (code, name, description, vegan, vegetarian)

RecipeIngredient (userID, recipeCode, ingredientCode, recipeTitle, ingredientName, quantity)

Keys userID, recipeCode in RecipeIngredient are foreign keys referencing Recipe, ingredientCode is foreign key referencing Ingredient, Each user has a single email and username. If a recipe or an ingredient are vegan, then they are also vegetarian.

Is the above database well designed (Meaning in BCNF)? If yes, say so. If not, make it in BCNF.

Q. 7 (3 pts) Consider a relation $R[A, B, C, D, E]$ and the FDs $AB \rightarrow C$, $AC \rightarrow DE$ and $A \rightarrow B$. Is it in BCNF ? (explain why). If not in BCNF, give a BCNF decomposition.

Q. 8 (4 pts) Given the relations $R(A, B, C, D, E)$ and $S(F, G, H)$, for each of the following functional dependencies, say if they are **violated** in this instance of the database.

A	B	C	D	E
1	1	1	3	7
2	2	2	4	7
2	3	2	5	7
2	4	2	6	7

F	G	H
1	1	1
1	1	1

$AD \rightarrow C$ ☐ YES ☐ NO
 $C \rightarrow DE$ ☐ YES ☐ NO
 $A \rightarrow E$ ☐ YES ☐ NO
 $FG \rightarrow H$ ☐ YES ☐ NO

Q. 9 (2 pts) If the relation R is in BCNF then it cannot contain duplicate values within any column

A. ☐ TRUE B. ☐ FALSE

Q. 10 (2 pts) If the attribute A for the relation R is Foreign Key referencing Y in the relation S, then it holds that Y in S can contain only values that appear in A for R.

A. ☐ TRUE B. ☐ FALSE

Q. 11 (4 pts) Consider the following schema that describes cars, their owners and their drivers:

Person [Pid, Name, Owns]

Car [Cid, Value, Driver, Color]

Person.Owns is foreign key referencing Car.Cid and

Car.Driver is foreign key referencing Person.Pid

Here are the operators that are important to run fast.

- Insert a new Car in the database (this is very important to run fast.)
- Given a value V find the color of the cars that have a Value greater than V
- Given an Car id, delete the car from the Car table that has that id.

What indexes you would create to facilitate the above operation? Indicate them by filling up the table below.

	Index Type (Clustered or Unclustered)	Index Kind (B+tree or Hash)	Attributes (on which attributes to define the index)
1			
2			
3			
4			
5			
6			
7			

Q. 12 (4 pts) Consider a relation $R[A,B]$ and a Relation $S[B,C]$. The attribute B in S is a key and has a hash index on it. The lookup cost of a hash index is 2. The size of a tuple in S is t_s . R has 100 tuples that take space of N pages on the disk. The attribute $R.B$ and is a foreign key referencing $S.B$. A page on the disk has size P . Find the cost to run the query $\pi_{B,C}(R \bowtie S)$ and save the results on the disk.